Haley Aaron

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Summary of Qualifications

Stanford engineer with experience designing scalable ETL pipelines and cloud-native tooling for large structured datasets seeking to apply these skills to healthcare and mission-driven data products. I have experience leading initiatives to automate and optimize technical workflows involving climate science, multi-hazard probabilistic risk assessment, and associated financial exposure across a variety of asset classes. I have developed projects and tooling across a variety of programming languages and cloud-based compute platforms to streamline processes, handle large amounts of data, simplify user experiences, and perform validation and sensitivity studies of results.

Skills

Programming: Python (dask, zarr, xarray, numpy, pandas, scipy, scikit-learn), Julia, R, Git, Matlab, VBA, Tcl
Cloud/Data: Google Cloud, Kubernetes, Snowflake, BigQuery, Docker, GIS, Tableau

Education

9.2011-6.2017 **Stanford University, M.S. and B.S. Civil and Structural Engineering**

Experience

10.2021-07.2024 **BlackRock, Vice President** — **San Francisco, CA**
Aladdin Financial Engineering – Sustainability Analytics

- Designed and deployed scalable cloud-native data pipelines to transform 200+ TB of structured and semi-structured geospatial, economic, and climate model data into asset-level financial risk metrics, using Dask, xarray, and Python on GCP Kubernetes clusters
- Developed robust ETL pipelines and command-line tooling for scalable processing of large spatiotemporal climate datasets, enabling reproducible workflows and cutting compute time while expanding production metrics from 200 to 400+ in Aladdin
- Delivered risk assessments for a major global banking client, processing a 500k+ property MBS portfolio in under 8 weeks to quantify physical climate exposure at scale
- Performed sensitivity analysis and attribution model construction to tie changes in corporate company fundamentals as well as portfolio level risks to shifts in climate and exposure variables
- Constructed suite of visualizations for portfolio level analysis in the context of climate risk to provide risk and portfolio managers snapshots of areas over or underweight in exposure to climate change

7.2021-10.2021 **Rhodium Group,** *Senior Product Development Analyst* — **San Francisco, CA**

- Translated complex academic research and proprietary climate modeling IP into production-ready pipelines, processing 200+ TB of simulation, geospatial, and economic data into scalable asset-level metrics establishing the technical foundation for BlackRock's Aladdin physical climate risk platform post-acquisition

7.2017-6.2021 **Arup,** *Analyst* — **San Francisco, CA**

- Performed probabilistic risk modeling and data analysis to quantify impacts of man-made and natural hazards on the built environment and relevant populations, including performing portfolio level seismic and wind engineering as well as modeling COVID transmission
- Built internal automation tools using Python, MATLAB, Julia, Tcl, and VBA to automate seismic and wind hazard prediction and mitigation, as well as building and component design
- Analyzed and engineered design solutions using finite element analysis for high-profile and multi-million-dollar seismic building projects and protective design components including to mitigate blast, ballistic, and vehicular threats
- Collaborated cross-functionally across software, mechanical, electrical, geotechnical, and tunneling disciplines as well as externally with clients involved in commercial real estate, security, natural hazard data analysis, and risk and resilience

3.2013-9.2014 **Aurora Solar,** *Research Analyst* — **Palo Alto, CA**

- Built Python tooling to surface solar incentives by zip code; helped inform early product direction through user research with installers

Leadership/Awards

11.2022 **Heart of BlackRock Performance Award** — **BlackRock**
Awarded to team which displays excellent performance and emotional ownership across all business units

4.2018 **Office-Wide Award for Young Leader of the Year - Buildings** — **Arup**
Most promising junior engineer in Buildings cost center, San Francisco office

10.2013-10.2014 **Stanford Daily,** *Opinions Columnist* — **Stanford**

10.2015-6.2016 **ASCE Stanford Student Chapter,** *Officer* — **Stanford**

3.2012-6.2015 **Stanford Varsity Swim Team,** *Captain (2014-2015)* — **Stanford**
Helped lead team of 23 in NCAA and USA Swimming competitions. Competed at 2012 U.S. Olympic Trials, and 2011, 2012, 2013, and 2014 U.S. National Championships